Incremental Learning-Enabled Fault Diagnosis of Dynamic Systems: A Comprehensive Review

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***Abstract*—Effective fault diagnosis is crucial for maintaining the reliability and safety of industrial systems. Incremental learning, which enables models to continuously update and adapt to new data or emerging fault classes without complete retraining, has recently gained attention as a promising solution for addressing nonstationary data streams in fault diagnosis applications. Nevertheless, most existing review articles on fault diagnosis adopt a broad perspective, primarily discussing general techniques such as deep learning and transfer learning, without providing a dedicated focus on incremental learning strategies. To the best of our knowledge, it is the first review focusing specifically on incremental learning-enabled fault diagnosis methods. In this work, state-of-the-art incremental learning-enabled fault diagnosis are systematically reviewed. These methods are categorized into distinct groups based on their incremental learning strategies and application contexts. In addition, major challenges associated with applying incremental learning to fault diagnosis, including concept drift and catastrophic forgetting, are discussed, along with emerging solutions proposed to address these issues. A novel taxonomy and perspective on incremental learning-enabled fault diagnosis approaches is presented, providing a timely and comprehensive reference for researchers and practitioners in this evolving field.**

***Index Terms*—Artificial intelligence, dynamic systems, fault diagnosis, incremental learning.**

I. Introduction

DYNAMIC systems play a crucial role in various industrial applications such as aerospace, manufacturing, and energy [1], [2], [3]. With the rapid development of modern systems, these systems are becoming increasingly complex. Effective fault diagnosis methods are essential for timely identification of potential faults, prevention of system failures, enhancement of system reliability and safety, avoidance of accidents, and protection of personnel and equipment. Consequently, fault diagnosis-based maintenance methods have garnered increasing attention in recent years [4], [5], [6], [7], [8], [9], [10].

Modern fault diagnosis methods are typically categorized into *knowledge-driven*, *model-driven*, and *data-driven* approaches [11], [12], [13]. Knowledge-driven methods rely on the expertise accumulated by domain experts over long-term practice to establish a knowledge base that simulate reasoning and decision-making processes of human experts [14], [15], [16], [17]. Model-driven methods use precise mathematical models of the system to generate residual signals that reflect the differences between expected behavior and actual operation [18], [19], [20], [21]. Fault diagnosis is then conducted based on the analysis of these residual signals. Although these methods employ different technologies and serve various purposes, all fault diagnosis systems share a common feature: the implementation needs to process measurements obtained during system operation. However, the effectiveness of existing methods often depends on a comprehensive understanding of system models, which limits their applicability [22].

Over the past few decades, data-driven fault diagnosis methods have been gaining increasing attention from both academia and industry [23]. These methods analyze and process operational data to perform fault diagnosis without requiring precise analytical models of the system. The evolution from expert experience to fully automated or intelligent systems has driven rapid development in data-driven approaches, including multivariate statistical analysis, signal processing, and machine learning methods. Due to the rapid advancements in artificial intelligence since the 21st century, machine learning methods with deep architecture have demonstrated significant advantages in fault diagnosis for complex systems, owing to their powerful feature extraction and pattern recognition capabilities. Consequently, these methods have been proven to achieve remarkable results in specific diagnostic tasks [24].

Traditional-learning-based fault diagnosis methods generally rely on constructing diagnostic models offline and then deploying them directly in online environments. The core of these approaches lies in how to *more accurately construct the mapping between fault data and fault modes during the offline training phase*, ensuring diagnostic accuracy. However, existing methods often fail to adequately account for the *dynamic nature* of real-world systems. These limitations not only pose several implementation constraints in practical applications but also create a significant gap between academic research and industrial deployments.

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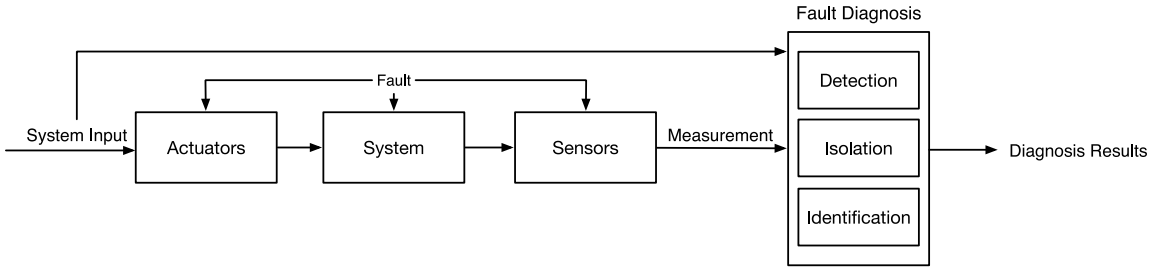


Fig. 1. General framework of fault diagnosis process.

Due to the complexity of fault scenarios and the unpredictability of degradation processes, obtaining comprehensive fault information during the offline phase remains a significant challenge. Fault data collected in the offline phase often fail to cover the full range of actual operating conditions [25], leading to incomplete or biased training datasets. In particular, industrial environments are subject to a variety of evolving factors, such as equipment aging, load variation, maintenance operations, and environmental changes (e.g., temperature, humidity, and dust), all of which can cause *shifts in the underlying data distribution* over time. These distributional shifts imply that the fault patterns observed during the offline phase may no longer fully represent the conditions encountered during real operations, resulting in model performance degradation if adaptation is not performed. In addition, the emergence of *novel fault modes* further exacerbates this challenge. Novel fault modes refer to new or previously unseen types of faults that were not captured during the initial training phase. These faults may arise due to equipment design modifications, unexpected interactions between system components, or the introduction of new operational processes. Since offline-trained models are inherently limited to recognizing known fault categories, they typically lack the capacity to detect or correctly classify these novel faults. Consequently, without online model updates or incremental learning mechanisms, such models may fail to provide timely warnings for critical but previously unencountered failure modes, potentially leading to severe safety risks or operational losses.

To address these challenges, diagnostic methods must possess higher dynamic adaptability when handling complex dynamic environments, such as unknown faults or long-term degradation effects. Recent research has explored incremental-learning-based fault diagnosis (ILFD) approaches. These methods dynamically adapt models to accommodate new fault patterns or system changes without requiring complete retraining. ILFD methods can progressively learn new operating conditions and fault characteristics, improving the model's adaptability to complex dynamic environments. Although still in the early stages of development, these approaches show promising potential in addressing challenges related to fault data scarcity, dynamic operating conditions, and model robustness.

In this article, we aim to provide a comprehensive overview of these evolving domains. We first review the development history, conceptual definitions, and technical taxonomies of fault diagnosis methods and incremental learning techniques. By examining the distinctions and interconnections between *batch-learning-based fault diagnosis* (BLFD) approaches and ILFD approaches, the unique advantages and challenges associated with each paradigm are highlighted. We then thoroughly review recent advancements in the literature from two perspectives, presenting a more comprehensive understanding of the state-of-the-art developments. The related research trends are further discussed and analyzed from three distinct perspectives. The challenges that ILFD methods encounter in representative application scenarios are also analyzed.

II. Basic Knowledge

A. Brief Concepts of Fault Diagnosis

Fault diagnosis techniques for dynamic systems are crucial methods for enhancing system reliability and reducing the risk of accidents. According to the official statement endorsed by the International Federation of Automatic Control (IFAC), a fault is defined as *an unpermitted deviation of at least one characteristic property or parameter of the system form the acceptable/usual/standard condition* [26]. Traditionally, a comprehensive fault diagnosis process must include the tasks of *fault detection*, *fault isolation*, and *fault identification* within a dynamic system, which can be described as [27] and [28].

1) *Fault Detection:* Determination of the faults present in a system and the time of detection.
2) *Fault Isolation:* Determination of the kind, location and time of detection of a fault, following fault detection.
3) *Fault Identification:* Determination of the size and time-variant behavior of a fault, following fault isolation.

Since the 1970s, a framework oriented toward problem-solving has been preliminarily established. Classic model-driven fault diagnosis methods typically proceed in the aforementioned three steps, thereby yielding specific analytical results regarding faults. The general framework of fault diagnosis process can be shown in Fig. 1. Model-driven fault diagnosis methods assume that the input/output (I/O) of a system conforms to a specific model structure,[1] such as a state-space model, with known parameters. Over the past few decades, these methods have garnered significant attention, particularly for linear systems. They have been fully integrated into vehicle control systems, robots, transport systems, power systems, manufacturing processes and process control systems [29]. However, they typically necessitate the

[1]Note: It refers to the input and output of the control system instead of the input and output data of the fault diagnosis model.

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precise identification of system model parameters, which poses challenges for fault diagnosis in more complex systems.

With the rapid development of artificial intelligence technology in recent years, some researchers have attempted to directly establish *nonlinear mappings* from system measurements to fault patterns. These approaches tend to emphasize fault detection and fault isolation tasks. Taking chemical systems as an example, classical fault diagnosis methods typically require a thorough understanding of the system model to analyze dynamic relationships and describe how the state of the system changes over time, forming the basis for effective diagnosis. However, for large and complex systems, precise modeling is often impractical. Consequently, data-driven fault diagnosis methods have gained significant attention as they can effectively overcome these limitations [24]. Hence, in recent years, an increasing number of fault diagnosis studies have gradually evolved from the logic of first fault detection and then fault isolation to directly classifying fault modes. In other words, these methods employ *fault classification* tasks to describe fault diagnosis and achieve the classical fault detection and isolation (FDI) tasks.

B. Different Perspectives of ILFD Approaches

Traditional machine learning paradigms usually operate in a *batch learning* manner, where a complete collection of training data is needed in advance to train the learner by following some learning algorithm. Such a paradigm requires the entire training data set to be provided before learning the task, and the training process is usually completed in an offline environment due to the expensive training cost. As discussed in [30] and [31], traditional batch learning methods have several limitations: 1) *low time and space efficiency* and 2) *poor scalability for large-scale applications*, as the learner typically needs to be retrained from scratch for new training data. In this case, although the traditional batch learning method has achieved good results in some fault diagnosis tasks, significant progress is still required before it can be widely adopted in practical scenarios.

In contrast to batch learning algorithms, incremental learning is a branch of machine learning on data that arrives sequentially, where the goal of learner is to learn and update the best predictor for future data incrementally [32]. Incremental learning overcomes the shortcomings of batch learning because the predictive learner can be updated with any new data. Therefore, taking into account the characteristics of system measurement in industrial applications, the idea of incremental learning is more suitable for industrial application scenarios and has been used in many specific applications in fields such as predictive maintenance.

In the context of fault diagnosis, methods across various scenarios are organized according to the structure illustrated in Fig. 2. Most existing machine-learning-based fault diagnosis methods are batch learning-motivated, assuming that the initial data used to construct the diagnostic model is complete. In the offline stage, diagnostic models can utilize the sufficient data to perform the training process. When the initial training data is incomplete, incremental learning methods can update the diagnostic models to handle more complex tasks. Based on the specific time of updates, these methods can be further divided into *incremental-online learning (OL)*-based FD approaches and *incremental-offline-learning*-based FD approaches. The former typically performs real-time updates in the online stage. Ideally, the deployment process should be completed before the arrival of system measurements. The update process of the latter usually occurs in the offline stage.

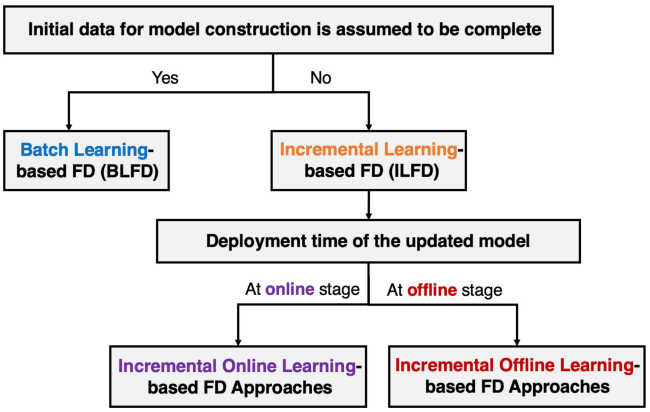


Fig. 2. Taxonomy of fault diagnosis methods based on data completeness and model update deployment time.

Remark 1: In the existing literature, incremental OL methods are often simplified to OL, which typically assumes that data arrives in a data stream, either one by one or batch by batch, and updates the diagnostic models after receiving each sample. The update process is required to be real-time. In contrast, incremental offline learning typically assumes that updates can be done offline after accumulating a certain amount of data and then redeployed in the online stage. The update process of such methods usually has higher complexity. Continual learning is typically regarded as a representative method that focuses on addressing the potential issue of *catastrophic forgetting* during the incremental learning process. For more information, please refer to [30], [31], [32], [33], and [34].

In general, the performance of learning-based fault diagnosis approaches can be significantly impacted by distribution drift, with incremental learning offering advantages over traditional batch learning in adapting to such changes. A schematic comparing ILFD and BLFD is shown in Fig. 3. In an ideal scenario, the data distribution during the operational phase should adhere to the *independent and identically distributed* (i.i.d.) assumption. The ideal performance curve is depicted by the red line. When a distribution drift occurs, indicating a change in data distribution, the performance of BLFD methods declines significantly. Conversely, ILFD methods aim to update the model to adapt to the new data distribution, thereby improving performance.

While incremental learning can be implemented in both online and offline settings, these two approaches differ significantly in their model update mechanisms and their applicability to fault diagnosis. Incremental-OL-based fault diagnosis approaches update the diagnostic model in real-time as new data arrives, making them particularly suited

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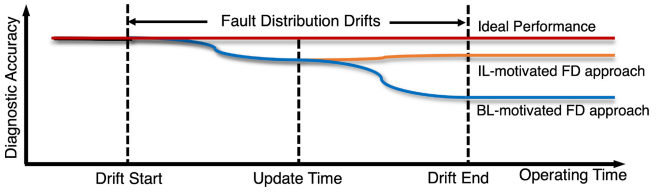


Fig. 3. Schematic illustrating the superior adaptability of ILFD approaches compared to BLFD approaches under fault distribution drifts.

for fast-changing environments. The diagnostic model needs to be deployed in real time, which is essential in scenarios such as real-time fault monitoring, where timely detection and response are crucial. These approaches are ideal for applications like machinery health monitoring or industrial systems operating under varying conditions, where faults must be detected immediately as they arise. By continuously updating the model with each incoming data point, it remains relevant to the latest operational state, providing the flexibility needed in environments with unpredictable changes or disruptions.

In contrast, incremental-offline-learning-based fault diagnosis approaches require the accumulation of data over time and perform model updates in batches. The diagnostic model should be deployed once conditions are met, such as when sufficient understanding of fault information under new operating modes has been achieved. Generally, these methods do not require specific timing for deployment. In such cases, fault diagnosis tasks benefit from enhanced stability and accuracy. These approaches are well-suited for applications where fault diagnosis can be performed after sufficient data has been gathered, such as in situations where fault patterns are complex and require extensive analysis. Incremental offline learning ensures a more thorough analysis of the data, leading to stable and precise model updates, which are particularly advantageous in scenarios where a deeper understanding of fault conditions under various operating modes is required.

III. Formulation and Motivation

In this section, some issues are first formulated. Subsequently, the general research motivations for ILFD approaches are analyzed.

A. Problem Formulations

For the sake of clarity in subsequent discussions, we provide a general formal definition in this section. As shown in Fig. 1, there may be component faults, actuator faults, and sensor faults typically. For linear dynamic system, the system model can be denoted as

$$z(k+1) = Az(k) + Bu_f(k) + w(k) + f_a(k) \tag{1}$$
$$m(k) = Cz(k) + Du_f(k) + v(k) \tag{2}$$

where A, B, C and D are system matrices, which have appropriate dimensions. Let $r_z, r_m \in \mathbb{Z}^+$ represents the dimensions of state variables and measurement variables, respectively. $z \in \mathbb{R}^{r_z}$, and $m \in \mathbb{R}^{r_m}$ are the system state and measurement. $w \in \mathbb{R}^{r_z}$ and $v \in \mathbb{R}^{r_m}$ are unknown noises. $f_a(k)$ represents the component fault of the system. The control input under actuator fault is denoted as $u_f(k)$, which can be modeled as

$$u_f(k) = \eta(k)u(k) + \xi(k) \tag{3}$$

where $\eta(k)$ represents the *multiplicative* actuator fault, and $\xi(k)$ represents the *additive* actuator fault. r_z, $r_m \geq 1$ are integers, which denotes the number of dimensions.

For nonlinear dynamic system, the system model can be denoted as

$$z(k+1) = \Phi\big(z(k), u_f(k), w(k), f_a(k)\big) \tag{4}$$
$$m(k) = \Psi\big(z(k), u_f(k), v(k)\big) \tag{5}$$

where $\Phi(\cdot)$ and $\Psi(\cdot)$ denote the nonlinear operator. u_f is also modeled as the form of (3).

Remark 2: It is worth noting that existing data-driven fault diagnosis methods of dynamic systems typically overlook the issues of sensor faults. Therefore, such issues are not considered generally for the construction of system model as shown in (1) and (4).

Consistent with the majority of fault diagnosis methods based on machine learning, the ILFD process typically requires obtaining fault information corresponding to the measurements in advance. In the literature, it is often referred to as the set of *fault label*. In the offline phase, measurements corresponding to different system states can be collected to form a training dataset $\mathcal{D}^{\text{off}}$. Assume that there are a total of N^{off} measurements and M^{off} kinds of fault modes $\mathcal{Y}^{\text{off}} = \{y_1, y_2, \ldots, y_{M^{\text{off}}}\}$. To maintain consistency with existing literature, the variable $\mathcal{X}$ is used to represent measurements when referring to samples in the following text. In this case, $\mathcal{D}^{\text{off}}$ can be represented as

$$\mathcal{D}^{\text{off}} = \left\{\left(\mathcal{X}_i^{\text{off}}, \mathcal{Y}_i^{\text{off}}\right)\right\}_{i=1}^{N^{\text{off}}} \tag{6}$$

where $\mathcal{X}_i^{\text{off}} \in \mathbb{R}^{r_m}$ and $\mathcal{Y}_i^{\text{off}}$ denote the ith instance and its fault mode in $\mathcal{D}^{\text{off}}$, respectively.

Typically, the diagnostic model Θ can be constructed during the offline phase, denoted as

$$\Theta := \mathcal{X}^{\text{off}} \to \mathcal{Y}^{\text{off}}. \tag{7}$$

During the online phase, the diagnostic model needs to predict the fault modes of the dynamic system based on the real-time measurements. Denote the sample collected at time t as $\mathcal{X}_t^{\text{on}}$. The predicted diagnostic result can be expressed as

$$\widehat{\mathcal{Y}}_t^{\text{on}} = \Theta\big(x_t^{\text{on}}; \theta\big) \tag{8}$$

where θ represents the model parameters of Θ. As discussed in Section II-B, ILFD typically attempts to perform incremental updates during either the offline or online phase. The data used for updates can be represented as

$$\mathcal{D}^{\text{new}} = \left\{\left(\mathcal{X}_i^{\text{new}}, \mathcal{Y}_i^{\text{new}}\right)\right\}_{i=1}^{N^{\text{new}}} \tag{9}$$

where $N^{\text{new}} \in \mathbb{Z}^+$ denotes the number of samples used for the update of diagnosis model. In other words, the diagnosis model can be updated during the offline stage or online stage via the following expression:

$$\theta^{\text{new}} = \zeta\big(\theta, \mathcal{D}^{\text{new}}\big) \tag{10}$$

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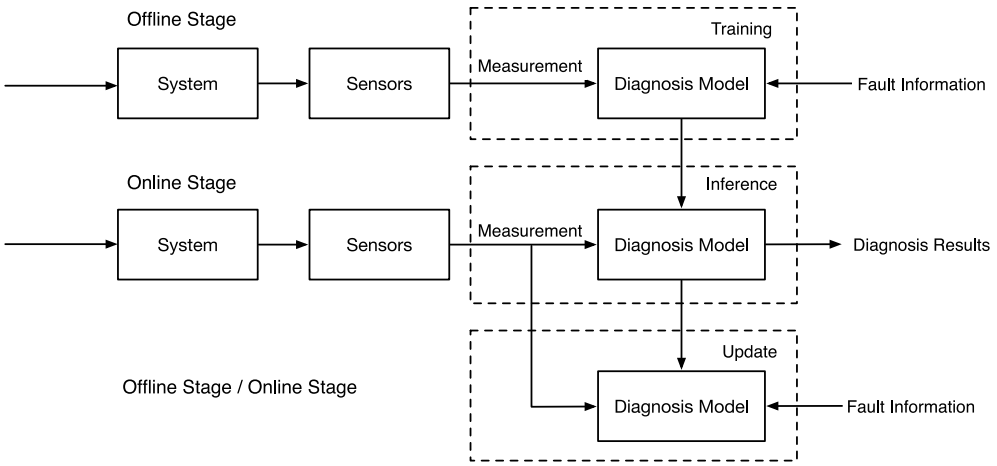


Fig. 4. General work-flow of ILFD process.

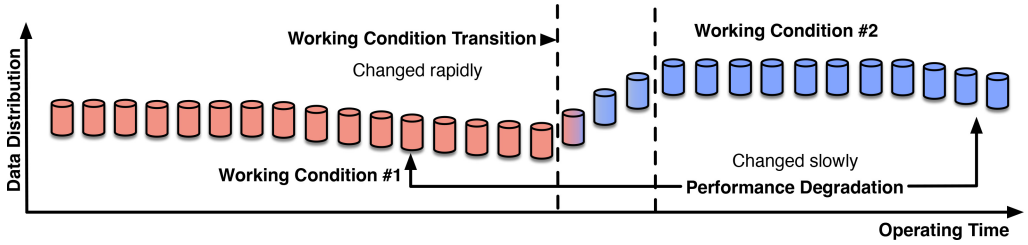


Fig. 5. Schematic of data distribution changes due to performance degradation and operating condition transitions.

where $\zeta(\cdot)$ denotes the update function. Its specific formulation is influenced by various factors such as the stage of update (offline or online) and the architecture of the diagnosis model. The main purpose is to minimize the expected risks after incremental updates, which can be defined as

$$\mathcal{R}(\theta) = \mathbb{E}_{(x,y)\ \sim(\mathcal{X}^{\text{on}},\mathcal{Y}^{\text{on}})}\left[\ell\left(\theta^{\text{new}}; x, y\right)\right] \tag{11}$$

where $\ell(\theta^{\text{new}}; x, y)$ represents the prediction error or loss of the diagnosis model given instance and its fault mode.

Considering the primary characteristics, the general work-flow of ILFD process can be summarized as Fig. 4.

B. Motivation Analysis

In practice, data are continuously generated and updated, with historical samples accumulating over time. During the early stages of equipment life cycles, the number of faulty devices is typically much smaller than the number of normal devices. As a result, the statistical features and patterns extracted from faulty devices may be incomplete, leading to potentially incorrect diagnostic results [35], [36]. Moreover, the initial understanding of potential system faults may be inadequate. If diagnostic models rely solely on samples accumulated during the offline phase for inference, their understanding of fault characteristics may be biased [37]. Capturing system variable information under different operating modes or environments involves high costs and practical limitations. Rare fault events are particularly difficult to capture, and system dynamics are often incompletely represented. Therefore, obtaining comprehensive fault information during the offline phase remains a significant challenge.

In addition, the dynamic nature of systems presents further challenges. Fault data collected during the offline phase often fail to cover the full spectrum of actual operating conditions. These data are typically sourced from controlled laboratory environments, which differ substantially from real industrial settings. Traditional machine learning models, which typically assume i.i.d., struggle in dynamic systems where such assumptions are rarely valid. As shown in Fig. 5, drifts in fault distributions can occur due to changes in operating conditions, environmental factors, and degradation effects, further complicating fault diagnosis [38]. In dynamic environments, both natural variations in operating conditions (such as adjustments in process parameters or fluctuations in environmental conditions) and unknown faults often coexist. Distinguishing between these two factors is critical but challenging. Normal variations in system characteristics can manifest as nonlinear drifts or random fluctuations in performance indicators, which may be misidentified as faults, leading to high false positives [39]. Conversely, unknown faults may be obscured by these normal variations, resulting in false negatives. The distinction is not adequately explored in existing methods, which often rely on static threshold settings or statistical feature analysis. As a result, traditional batch learning methods, which cannot effectively handle real-time updates, fail to address the complexities posed by such dynamic conditions.

These challenges highlight the need for diagnostic models capable of real-time learning and updating. The existing literature has identified two key issues in this context: 1) *fault distribution drifts* and 2) *novel fault modes*, which require

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TABLE I
MAIN ISSUES TO BE CONSIDERED FOR ILFD PROCESS

Main Issues	Description	Formulation
fault distribution drifts	Data distribution of system measurements changes over time due to multiple factors.	$\Theta : \mathcal{X} \times C \rightarrow \mathcal{Y}$
novel fault modes	Exist new fault modes that were not present at the training stage as the process evolve.	$\Theta : \mathcal{X} \rightarrow \mathcal{Y} \times C$

Notes: C represents the incremental changes in the measurements $\mathcal{X}$ and the fault information $\mathcal{Y}$, indicating variations that dynamically adjust the behavior of both components.

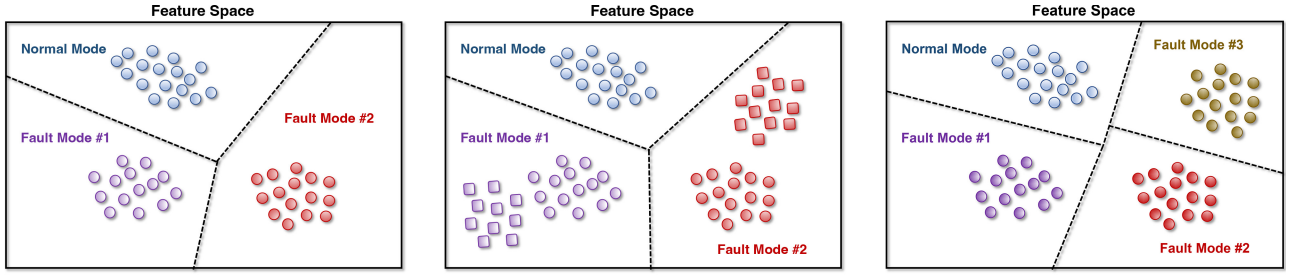


Fig. 6. Conceptual illustration of fault distribution drifts and novel fault modes in feature space. The dashed line represents the decision boundary. (a) Original diagnostic tasks. (b) Fault distribution drifts. (c) Novel fault modes.

more focused attention for effective fault diagnosis. A conceptual illustration of these two issues in feature space is shown in Fig. 6, while detailed descriptions of these issues are summarized in Table I.

On the one hand, dynamic systems often operate in nonstationary environments due to factors such as operational condition changes and performance degradation [40], [41], [42], causing the distribution of fault samples to change to some extent. A key phenomenon known as *concept drift* is often observed [43]. In practice, most cases involve mixed drift types, meaning that multiple types of drifts can occur simultaneously [44], [45]. In mechanical systems, operation of equipment may lead to changes in fault patterns and frequencies due to the wear and aging of components, which alters data distributions of fault data. In chemical systems, adjustments in process parameters, variations in raw materials, or fluctuations in environmental conditions can change the operational state of the system, thereby affecting fault distribution. It is primarily due to changes in reaction processes and equipment performance under different conditions. In real-life scenarios, the occurrence of the above drifts can lead to changes in the distribution of fault data, thereby affecting the real-time performance of machine-learning-based fault diagnosis methods. Incremental learning methods enable diagnostic models to update in real-time, adapt to fault distribution changes, thereby improving system reliability and stability, reducing false positives and false negatives, optimizing maintenance schedules, and lowering equipment maintenance costs.

On the other hand, the emergence of novel fault modes poses significant challenges to traditional FD methods. The application of new materials, new technologies, and continuously changing operating environments can result in previously unseen fault modes [46], [47], [48]. In mechanical systems, the application of new materials and manufacturing technologies may introduce new fault modes due to the diversity and unpredictability of faults associated with new material properties and the complexity of new technologies. In chemical systems, the introduction of new chemical processes or catalysts can lead to the emergence of new fault modes, as these new processes and catalysts may reveal unknown fault risks during their initial operation. If diagnostic models cannot promptly identify and adapt to these novel fault modes, severe faults may go undetected, compromising system safety and stability. Incremental learning methods enable diagnostic models to quickly learn and update when new fault modes appear, which enhances the flexibility and accuracy. As a result, they provide comprehensive monitoring of equipment health and help prevent potential major faults.

IV. RECENT PROGRESS FOR INCREMENTAL-ONLINE-LEARNING-BASED FD APPROACHES

As the volume of data increases, incremental-OL-based FD approaches need to be capable of performing rapid updates during the online phase without compromising real-time performance, especially in dynamic environments. To address the challenges outlined in Section III-B, the following characteristics must be thoroughly considered.

1) *Rapid Update Speed With Real-Time Deployment:* Considering that diagnostic models are required for real-time deployment, they should be capable of rapidly learning and updating as new data arrive. In this context, the update speed of such diagnostic models is typically in higher demand [49], [50], [51].
2) *Efficient Update Capability With Minimal Monitoring Data:* The method should update the model effectively with a minimal amount of new samples, minimizing reliance on large datasets while ensuring

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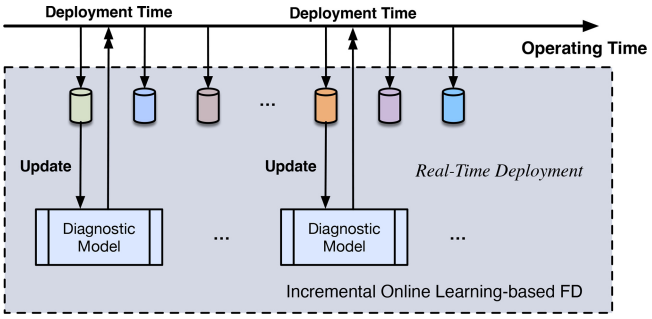


Fig. 7. Schematic of incremental OL-based FD approaches.

performance without requiring extensive computational resources [52], [53], [54].

3) *Effective Utilization of Temporal Relationships:* Diagnostic models should effectively utilize temporal relationships to capture the dynamic behavior and trends of the system over time, thereby enhancing the accuracy and robustness of the diagnostic model [55], [56].

These characteristics enable such diagnostic methods to achieve efficient incremental updates during the online phase, making them a core focus in the design of methods in existing literature. Therefore, incremental-OL-based FD methods can significantly enhance the adaptability, flexibility, and accuracy of diagnostic systems, providing robust support for the reliable operation of dynamic systems. In this context, incremental-offline-learning-based FD methods have been impressively developed over the past few years. A schematic of incremental-OL-based FD approaches is illustrated in Fig. 7 to provide a clearer overview of their structure and key components.

A. Approaches for Handling Fault Distribution Drifts

To effectively address the challenges introduced by fault distribution drifts, diagnostic models are required to perform timely and effective updates. A key concern is: *how to design model architectures to accommodate these updates*, which has been a subject of extensive discussion.

In light of the need for incremental-OL-based diagnosis methods to promptly respond during the current phase, the structure of *shallow learning* techniques is often employed as the primary framework for designing update strategies at the early stages of research. To address the limitations of traditional local linear embedding (LLE) in mechanical fault diagnosis, an incremental supervised LLE (I-SLLE) was introduced for fault diagnosis in submersible plunger pumps [57]. The I-SLLE algorithm employs matrix decomposition techniques to manage out-of-sample data. It updates subsets of old low-dimensional coordinates to enhance precision and is achieved through iterative updates applied to all data. Subsequently, the test data is classified in feature space using *support vector machines (SVMs)*. Similarly, an incremental enhanced supervised LLE is also introduced for nonlinear dimensionality reduction, effectively embedding data into low-dimensional manifold structures [58]. The low-dimensional representations of new fault samples are then fed into an adaptive nearest neighbor classifier. It identifies fault modes by selecting the nearest training samples based on representational distance and employs a weighted strategy to enhance accuracy.

An improved incremental *Gaussian naive bayes* (I-IGNB) learning algorithm has been employed to enhance model update efficiency by updating key parameters such as prior and conditional probabilities [59]. Incremental SVMs have also been considered for more effective adaptation to new data [60], [61], [62]. A multikernel SVM that incorporates incremental learning was proposed. It aims to improve accuracy and reduce maintenance costs and has been successfully validated on two mass-produced complex industrial boards. Compared to single-kernel SVMs and artificial neural network methods, its superiority has been well validated [63]. In addition, some methods attempt to use shallow learning techniques for diagnostics, which involve detection and isolation [64], [65].

In recent decades, learning architectures based on randomized neural network (RNNet) structures, represented by random vector functional-link neural network (RVFL) [66], extreme learning machine (ELM) [67], and broad learning system (BLS) [68], have been used to design diagnostic model architectures considering their good incremental update capabilities [69], [70], [71]. These methods, although being part of neural network structures, often perform incremental updates based on least squares method, avoiding the time costs associated with parameter tuning in optimization processes via back-propagation techniques. For instance, Mao et al. introduced an online sequence ELM prediction method considering the smaller quantity of fault data collected online compared to normal data. During the online phase, granules and main curves are reestablished for sequentially arriving bearing data. A balanced sample set is formed through oversampling and under-sampling processes to dynamically update the diagnostic model [72].

RNNet-based diagnostic methods have seen rapid development in recent years and have been applied to address additional problems such as time-varying operating conditions and performance degradation [73], [74], [75]. An analytical incremental learning (AIL) method based on ELM has been proposed to address performance degradation issues in early fault diagnosis of electric power transformers [76]. A BP-incremental BLS, designed to cope with challenges brought by nonstationary vibration signals, has been introduced to speed up model updates [77]. A novel method for real-time multimode fault diagnosis using evidential ensemble preference was also developed, guiding the identification process of operating conditions and utilizing an incremental BLS for diagnostics, achieving excellent diagnostic results [73]. To address incremental updates between multiple operating conditions, a new cross-domain class-incremental broad network (CDCIBN) has been proposed to handle diagnostic problems of nonindependent and identically distributed continuous stream data [78]. In [55], a latent variable-guided BLS (LVGBLS) was proposed to construct the diagnostic model. An incremental update procedure was then developed using pseudo-label learning, designed to adapt to dynamic process changes while minimizing the labeling costs. These methods

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fully leverage the rapid updating capabilities of the randomized structure, achieving impressive performance across multiple tasks.

Although the aforementioned methods have achieved success in certain applications, their generalization still require further enhancement for more complex diagnostic tasks. In this context, the concept of ensemble learning has been considered in the literature. There is significant evidence that ensemble learning can effectively and flexibly deal with changes in data patterns in nonstationary environments [79], [80]. The primary reason is that ensemble learning methods can flexibly adjust the weight allocation to weak learners, thereby rapidly changing the understanding of the ensemble learners. Consequently, several designs focused on addressing fault distribution drifts have been introduced.

For instance, Zhang et al. [81] proposed a method based on a multivariate ensemble-based incremental SVM, which uses existing model parameters to learn new fault information gradually without needing previous monitoring data. It effectively detects various faults, including complex faults and faults of different severity levels, by linking multiple monitoring variables with their corresponding defect types. Similarly, Jing and Zhenya [82] utilized semi-supervised learning to lessen the dependency on prior fault knowledge. Newly generated system measurements are first mapped to several clustered label vectors using kernel principal component analysis (KPCA). A diverse subset is then selected for incremental updates based on mutual information. In [83], the idea of online semi-supervised learning is introduced to make full use of fault information from offline phases. It reduces the dependency on fault information collected during online processes, which offers practical value. While in [84], an adaptive ensemble fault diagnosis method has been proposed to address two practical challenges in complex systems: 1) the lack of sufficient historical fault instances and 2) the absence of personalized features in the diagnosed devices. Hidden Markov models (HMMs) are constructed to extract patterns from historical fault and normal instances. Diagnosis made by supervised and unsupervised methods are then integrated through hidden states identified by HMMs, overcoming the challenge of insufficient historical fault instances. An interesting approach was proposed in [85]. The forms of data changes were discussed from the perspective of mathematics. A strategy that adapts to these changes based on the least risk was proposed. Following this strategy, an adaptive incremental diagnostic model was developed, which has proven effective in real industrial settings such as the Tennessee Eastman (TE) process and the multiphase flow facility (MPFF).

B. Approaches for Managing Novel Fault Modes

In addition, such incremental-OL-based FD approaches have been widely employed to address a critical issue is *how to effectively manage novel fault modes that emerge during the online phase*. Traditional BLFD approaches typically require the assumption that the understanding of fault modes is comprehensive. However, as described in Section III-B, novel fault modes often emerge in practical scenarios, potentially diminishing the diagnostic performance of classical methods. The majority of methods require an initial determination of whether the collected system measurements belong to known fault modes. Subsequently, specific strategies are designed to execute the update process. These methods generally include *novelty detection* modules for fault modes.

Over the past decade, considerable exploration has been conducted to address this issue. A common idea is to identify the fault direction of data corresponding to novel fault modes after detecting that current system measurements do not belong to known fault modes, thereby facilitating better fault isolation. Such methods typically aim to address FDI tasks, rather than regarding them as fault classification tasks. For instance, Yang et al. [86] proposed a novel class-incremental FDA scheme, which involved identifying novel fault modes from known ones based on partial F-values with cumulative percent variation (CPV). Once the F-direction is obtained, the cosine of the angle between the new fault samples and fault modes can be added to the database. Liu et al. [87] used a max-min deviation rule with online updating capability to determine whether new samples originated from novel fault modes. It was achieved by considering both the maximum class posterior probability and the entropy in a weighted sum. An online fault diagnosis method based on Bayesian incremental learning was also proposed in [88]. Due to the architectural advantages of FDA and naive Bayes, such methods can conveniently solve the aforementioned issues through incremental updates to prior probabilities without retaining all historical information. However, it is relatively difficult for such structures to deal with more complex tasks.

In [89], the incremental support vector data description (ISVDD) was used for rapid detection of novel fault modes in continuous equipment monitoring. An ELM with incremental output structure (IOELM) was then introduced, adapting to the recognition of new fault modes by adding output nodes. While Fu et al. [90] also utilized ISVDD to detect the novel fault modes. A broad autoencoder (BAE) with class-incremental learning capability was designed to adapt to new incoming fault samples and novel fault modes. Regarding unknown fault modes, Shi et al. [91] further addressed the issue of limited amount of fault sample with the proposed *semi-supervised class-incremental broad network (SSCIBN)*. Benefiting from the structural advantages of RNNet, a class-incremental closed-form solution was used to address the updating issues of diagnostic models under novel fault modes in [92] and [93].

Similarly, the concept of ensemble learning has also been utilized to manage novel fault modes. For novel fault modes, the diagnostic capability of ensemble learners for such faults can be enhanced by adding new base learners. For example, Razavi-Far et al. [94], [95] introduced the Learn++ algorithm in OL for fault classification. Novel fault modes were identified by introducing an ensemble-based incremental diagnostic scheme. These algorithms were capable of detecting unseen fault modes in the monitored data stream, classifying unknown fault modes until their correct labels were assigned. Sankavaram et al. [96] investigated an age-based incremental learning methodology for ensemble classifier with evolving

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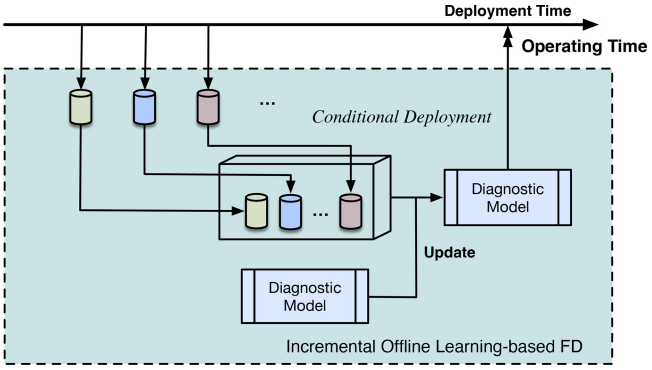


Fig. 8. Schematic of incremental-offline-learning-based FD approaches.

databases. Its effectiveness was validated on a dataset of automotive control subsystems.

V. Recent Progress for Incremental-Offline-Learning-Based FD Approaches

Another common paradigm is the incremental-offline-learning-based FD approach. These approaches typically focus on incremental updates at the task level, with the updated diagnostic model deployed in the online phase only when specific conditions are met. To address the challenges outlined in Section III-B, the following characteristics must be considered.

1) *Nonlinear Feature Extraction Capability:* Given the inherent complexity and variability of system behaviors, diagnostic models should be capable of extracting nonlinear features from the collected data, which is essential for capturing complex fault patterns and relationships.
2) *Richness of Accumulated Monitoring Data Information:* The accumulated monitoring data should be sufficiently rich and diverse to avoid cognitive limitations arising from insufficient information on fault characteristics, ensuring that the diagnostic model can generalize well and make accurate predictions under different operational conditions [97], [98], [99].
3) *Mitigation of Catastrophic Forgetting Effects:* The design of diagnostic model architectures should incorporate the ability to learn new information without forgetting previously acquired knowledge, thus preventing significant performance degradation after updates [100], [101].

The effectiveness of hand-crafted features typically requires engineers to have a thorough understanding of the operational processes. To better address these issues, deep learning techniques are commonly employed to design the model architecture. More complex fault features can be captured in an end-to-end manner, thereby avoiding the limitations of classical feature engineering in practical diagnostic tasks. The application of *continual learning* (also known as *lifelong learning*) technology has attracted widespread attention [102]. A schematic of incremental-OL-based FD approaches is illustrated in Fig. 8 to provide a clearer overview of their structure and key components.

A. Approaches for Handling Fault Distribution Drifts

In contrast to the incremental-OL-based FD methods mentioned in Section IV, some approaches aim to incrementally update the model during an offline phase after accumulating sufficiently rich historical data. The idea of the technical approach is also intuitive. If fault distribution drift occurs, the updated model can perform the diagnostic task effectively, addressing issues such as performance degradation and multi-condition fault diagnosis to some extent.

A common approach involves retraining key components of the diagnostic model with new samples obtained during the online phase to adapt to new fault patterns [103]. Carino et al. [104] proposed a novel detection method based on SVMs to identify previously unconsidered fault patterns. Once a new fault pattern is detected, the diagnostic model is retrained. Its effectiveness was validated on an automotive end-of-line test machine. Similarly, in [105], deep learning algorithms were employed to extract feature patterns from newly generated data. These new data features were compared with fault patterns extracted from historical data to adjust the weights corresponding to different fault patterns in the training set. Finally, SVM was used to perform supervised classification of the weighted patterns.

Given that deep neural network (DNN)-based diagnostic models typically have a large number of parameters, frequent retraining of these models is considered time-consuming. Therefore, addressing catastrophic forgetting in diagnostic models has emerged as an important research topic. The distribution of fault features can vary significantly under different load conditions, ambient temperatures, and other operational environments. Moreover, real-world devices often consist of multiple critical components, each exhibiting distinct fault characteristics. Consequently, diagnostic tasks evolve continuously over time. An ideal diagnostic model should be capable of continuously learning critical information from new tasks while preserving the knowledge acquired from previous tasks.

To address catastrophic forgetting in fault diagnosis tasks, various incremental learning strategies have been proposed in the existing literature. For instance, the pseudo-rehearsal method was proposed in [106] to mitigate forgetting by generating synthetic data to approximate past knowledge. However, the quality and authenticity of pseudo-data remain critical challenges. Sample memory strategies, like the central sample preservation method introduced by Min et al. [107], effectively retain important information but require substantial storage resources. Knowledge distillation (KD)-based FD approaches have also gained attention [108], [109], [110]. For example, Guan et al. [111] utilized adaptive KD and representative sample selection to handle performance degradation, while Zhu et al. [112] applied KD loss to preserve critical prior information. Furthermore, Hu et al. [113] proposed a dynamic weight correction algorithm to balance model stability and plasticity, albeit with increased computational complexity. In addition, dynamic network expansion techniques, such as dynamic weight aggregation [114], task-incremental learning strategies like TiBLS [115], and replay-enhanced methods like R-REMIND [116], have been explored to enhance the model's adaptability without catastrophic forgetting. More recently,

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strategies such as task-aware dynamic masking [117] and elastic expansion mechanisms [118] have been proposed to optimize learning flexibility.

Overall, while these approaches demonstrate strong potential in preserving prior diagnostic knowledge in evolving environments, challenges such as computational cost, storage demands, and performance tradeoffs between stability and performance remain active areas of research.

B. Approaches for Managing Novel Fault Modes

It is also critical to manage novel fault modes due to the dynamic and complex nature of industrial systems. Various approaches have been proposed in the literature to address this challenge [119], [120], [121], [122]. For instance, Zhang et al. [123] utilized sparse autoencoders to identify and generate new labels, allowing diagnostic models to adapt to and recognize novel fault modes. Similarly, an incremental hierarchical expansion of DNNs was proposed to handle continuously incoming novel fault modes [124]. By using fuzzy clustering to group similar faults into super-classes, their approach allows submodels to share the same architecture and train in parallel, facilitating efficient model updates. To balance stability and plasticity in fault diagnosis models, Chen et al. [125] introduced dual-branch aggregation networks. It employs aggregated weights to ensure that models retain old knowledge while learning new information, thus addressing the challenge of model failure due to the introduction of novel fault modes under complex operational conditions. Following the idea of meta-learning, Liu et al. [126] proposed a continual learning model based on weight space meta representation (WSMR) for class-incremental fault diagnosis of switch pumps. In [127], feature-enhanced continual learning was proposed to maintain the core structure of fault diagnosis models. By using an effective distillation strategy to remove redundant parameters and feature dimensions, such a method ensures that the diagnostic model remains efficient and accurate over time. The *replay-based* strategy has also been well considered in [128] and [129]. Liu et al. [129] focused on generative feature replay for lifelong learning, aiming to retain essential features for ongoing fault diagnosis, which complements the work in [127] to some extent by addressing the challenge of retaining critical information. Few-shot learning has emerged as another important research hotspot in recent years, both at the system-level and the component-level [130], [131], [132], [133], [134], [135].

In addition, a two-stage diagnostic model was also proposed in [136], which first clusters fault samples into distinct feature groups and then identifies fault modes based on the similarity between samples and these clusters. It effectively identifies novel fault modes using existing features, aligning with the continual learning strategies. Similarly, Liu and Wu [137] proposed a continual learning method based on down-sampling and model-agnostic meta-learning (CL-DMAML), which further highlights the importance of balancing new and existing knowledge of fault information. Huang et al. [138] combined multifidelity information fusion (MFIF-CIL) with class-incremental learning for rolling bearing faults, which aligns with the continual learning strategies of previous works by ensuring the model adapts to new fault modes while retaining essential knowledge. The works discussed in [139] and [140] tackled the issues of fault frequency differences and the need for complete model retraining.

VI. Research Trends for ILFD Approaches

Based on the above analysis, it is clear that ILFD methods have exhibited distinct research trends in learning architectures, main challenges, and system objects, which can be summarized as follows.

1) *Learning Architectures:* In recent years, incremental learning architectures have gradually shifted from traditional shallow learning and ensemble learning methods to diagnostic models with broader or deeper architectures. For incremental-offline-learning-based FD approaches, deep learning frameworks are becoming increasingly dominant. In particular, technologies such as lifelong learning and transfer learning have provided opportunities to address challenges related to novel fault modes and time-varying operating conditions [149]. In contrast, incremental-OL-based FD approaches remain predominantly reliant on diagnostic models with shallow architectures. These paradigms are generally lightweight, fast to train, and capable of real-time updates. As a result, an increasing number of approaches leverage these paradigms to address tasks with high demands on model deployment, such as performance degradation [55]. Compared to classical shallow learning or ensemble learning methods, RNNet-based shallow learning approaches are gaining attention for their ability to balance flexibility and model capacity.
2) *Main Challenges:* Over the past decade, research on ILFD has grown significantly, particularly since 2018. As application requirements have evolved, the focus of existing research has shifted from rapid adaptation to more complex challenges, such as effectively handling fault distribution drift and novel fault modes under time-varying operating conditions. These issues are particularly relevant in high-dimensional complex systems. Meanwhile, the problem of catastrophic forgetting has emerged as a significant issue. During the learning process, diagnostic models continuously incorporate new information, which can lead to the forgetting of previously acquired knowledge, thereby affecting long-term model performance. Maintaining a balance between model update efficiency, learning outcomes, and data diversity has become a key research focus.
3) *System Objects:* The application of ILFD has diversified significantly in recent years. Initially, incremental learning methods were primarily applied to fault mode recognition in mechanical systems. With the advancement of research, their scope has gradually expanded to include chemical and other complex systems. Furthermore, the rapid development of the industrial Internet and smart manufacturing has driven

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TABLE II
SUMMARY OF RECENT TYPICAL INCREMENTAL-LEARNING-BASED FD APPROACHES

Year	Literature	Architecture	Learning Paradigm	Major Challenges	System Object		
					Mechanical	Chemical	Others
2012	[94]	EL	incremental online learning	novel fault modes			✓
2014	[63]	SL	incremental online learning	fault distribution drifts			✓
2014	[89]	SL*	incremental online learning	novel fault modes	✓		
2015	[96]	EL	incremental online learning	novel fault modes			✓
2015	[81]	EL	incremental online learning	fault distribution drifts	✓		
2015	[141]	SL	incremental online learning	fault distribution drifts			✓
2016	[57]	SL	incremental online learning	fault distribution drifts	✓		
2017	[95]	SL*	incremental online learning	novel fault modes	✓		
2017	[82]	EL	incremental online learning	fault distribution drifts	✓		✓
2018	[60]	SL	incremental online learning	fault distribution drifts	✓		
2018	[123]	DL	incremental offline learning	novel fault modes	✓		
2018	[104]	DL	incremental offline learning	fault distribution drifts	✓		
2018	[105]	SL	incremental offline learning	fault distribution drifts	✓		
2018	[59]	SL	incremental offline learning	fault distribution drifts			✓
2019	[86]	SL	incremental online learning	novel fault modes		✓	
2019	[46]	SL	incremental online learning	novel fault modes		✓	
2019	[76]	SL*	incremental online learning	fault distribution drifts			✓
2019	[69]	SL*	incremental online learning	fault distribution drifts	✓		
2019	[124]	DL	incremental offline learning	novel fault modes		✓	
2020	[88]	SL	incremental online learning	novel fault modes			✓
2020	[142]	DL	incremental offline learning	fault distribution drifts		✓	
2021	[83]	SL*	incremental online learning	fault distribution drifts	✓	✓	
2021	[106]	DL	incremental offline learning	fault distribution drifts			✓
2021	[62]	SL	incremental online learning	fault distribution drifts	✓		
2022	[90]	SL*	incremental online learning	novel fault modes	✓		
2022	[85]	EL	incremental online learning	fault distribution drifts		✓	
2022	[125]	DL	incremental offline learning	novel fault modes	✓		
2022	[116]	DL	incremental offline learning	fault distribution drifts	✓		
2022	[115]	SL*	incremental offline learning	fault distribution drifts	✓		
2022	[139]	DL	incremental offline learning	novel fault modes		✓	
2022	[111]	DL	incremental offline learning	fault distribution drifts	✓		
2023	[92]	SL*	incremental online learning	novel fault modes		✓	
2023	[143]	SL*	incremental online learning	fault distribution drifts		✓	
2023	[140]	DL	incremental offline learning	novel fault modes	✓		
2023	[144]	DL	incremental offline learning	novel fault modes			✓
2023	[136]	DL	incremental offline learning	novel fault modes		✓	
2024	[93]	SL*	incremental online learning	novel fault modes	✓		
2024	[145]	SL*	incremental online learning	fault distribution drifts	✓		
2024	[91]	SL*	incremental online learning	novel fault modes	✓		
2024	[78]	SL*	incremental online learning	fault distribution drifts	✓		
2024	[127]	DL	incremental offline learning	novel fault modes	✓		
2024	[112]	DL	incremental offline learning	fault distribution drifts	✓		
2024	[146]	DL	incremental offline learning	fault distribution drifts	✓		
2025	[108]	DL	incremental offline learning	fault distribution drifts	✓		
2025	[147]	DL	incremental offline learning	novel fault modes			✓
2025	[148]	DL	incremental offline learning	fault distribution drifts			✓

Notes: SL denotes shallow learning; SL* denotes RNNet-based shallow learning; EL denotes ensemble learning; DL denotes deep learning.

these methods to shift from component-level applications to system-level applications. Related techniques are increasingly expected to integrate data from various sources and dimensions, enabling the processing of more complex fault modes, particularly those associated with system faults arising from intricate interactions and multifactor influences.

Remark 3: Although commonly used metrics such as accuracy and F1-score are widely adopted in fault diagnosis studies, there is currently no universally standardized evaluation framework, particularly for ILFD. Different studies select evaluation metrics based on application-specific needs, data characteristics, and experimental settings. In ILFD scenarios, time-sensitive indicators such as the real-time false alarm rate (RTFAR), real-time missed alarm rate (RTMAR), and cumulative averaged metrics (e.g., cumulative accuracy and cumulative F1-score) have been proposed to better capture model adaptability and long-term stability. However, these

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specialized metrics have not yet been universally standardized across the community.

VII. Typical Real-Word Applications

As shown in Table II, ILFD methods have been applied across a range of domains. In this section, we provide brief discussions of two representative application areas: 1) mechanical and 2) chemical systems.

A. Mechanical Systems

Mechanical fault diagnosis is of significant practical importance [150]. Faults in mechanical systems can lead to serious safety hazards, posing a threat to the personal safety of operators. In addition, they can cause disruptions in the production process, negatively affecting product quality. Mechanical systems typically comprise critical components such as bearings, gearboxes, rotors, and pumps [54], [151]. These systems are usually equipped with numerous sensors capable of collecting high-frequency data, such as vibration, temperature, pressure, and rotational speed [152]. The combination of high sampling rates and advanced feature extraction techniques enables a timely reflection of the system's operational state. Despite these capabilities, several challenges remain in the design of effective diagnostic models.

Performance degradation is a critical issue in the fault diagnosis of mechanical systems, as it directly impacts the effectiveness of evaluation methods developed during the offline stage [153]. Over time, changes in the stress and structure of system components can lead to functional failures, with the degradation process often being complex and highly unpredictable. As illustrated in Fig. 9, performance degradation typically manifests as gradual and continuous shifts in the distribution of monitoring data. Such challenges highlight the need for effective deployment and continual adaptation of diagnostic models during online operations. In this regard, incremental OL methods, which offer greater adaptability, are particularly well-suited to address performance degradation [55]. For example, He et al. [55] proposed an incremental update-based FD scheme that achieved up to 99.6% accuracy on the XJTU-SY bearing dataset under dynamic degradation conditions, which demonstrates the practical advantages of incremental learning in maintaining reliable diagnostic performance in evolving industrial environments.

B. Chemical Systems

Fault diagnosis in chemical processes has been a significant research focus for many years. Unlike faults in other industrial processes, those in chemical processes are more likely to result in severe accidents, such as toxic gas leaks and explosions [154], [155]. The rates of chemical reactions are often nonlinear, with many following the *Arrhenius* equation [156]. To maintain process parameters within desired ranges, feedback control systems are widely employed. However, these systems also introduce nonlinearities in their responses and control strategies. Due to the inherent difficulty in accurately modeling chemical processes, data-driven fault diagnosis methods are commonly used [157].

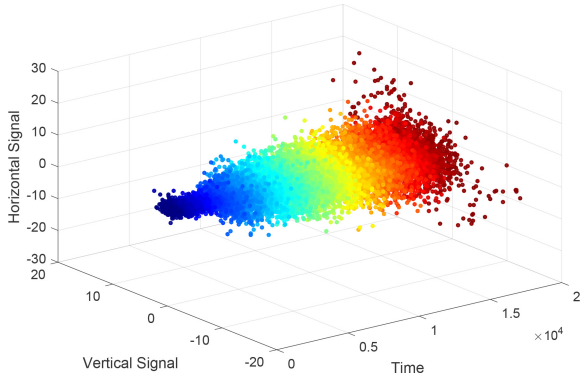


Fig. 9. Visualization of performance degradation [55]. The colors indicate that the bearing is in the process of degradation to different degree.

Modern chemical processes are typically equipped with numerous sensors that collect complex, high-dimensional, and sparse fault diagnosis data. Process parameters such as temperature, pressure, flow rate, and concentration are often interdependent, and any abnormal changes in these parameters can trigger faults. Unlike mechanical systems, the sampling frequency in chemical processes is generally lower, and actuator inputs typically have delayed effects, leading to response lags. The delay creates favorable conditions for incremental OL methods, which can rapidly adjust to changes in operating conditions, making them well-suited for adapting to seasonal variations and production adjustments.

Moreover, the operating modes of chemical processes are often difficult to control precisely. For instance, in blast furnace fault diagnosis, variations in coal powder quality affect combustion efficiency, which in turn influences operating conditions. Chemical processes often involve multistep and side reactions, complicating the prediction of novel fault modes. As shown in Fig. 10, time-varying transition conditions are often unavoidable, and transition condition data during the offline phase are typically unavailable due to operational differences. In such cases, incremental updating strategies are crucial for adjusting diagnostic models in real-time. Incremental-offline-learning-based fault diagnosis methods ensure diagnostic accuracy by continually learning from task variations across different operating conditions.

VIII. Conclusion and Prospects

In this review, we have first briefly reviewed the development history, conceptual definitions, and technical taxonomies of fault diagnosis methods and incremental learning techniques. We have discussed in detail the differences and connections between BLFD methods and ILFD approaches. Subsequently, we have thoroughly reviewed the recent developments in the existing literature from two perspectives. The related research trends have been further discussed and analyzed from three distinct perspectives. We have also analyzed the challenges that ILFD methods should face in representative application scenarios. Of note, we apologize for any omissions due to the continuous emergence of new publications.

Based on the review of existing literature, we summarize potential directions for future research as follows.

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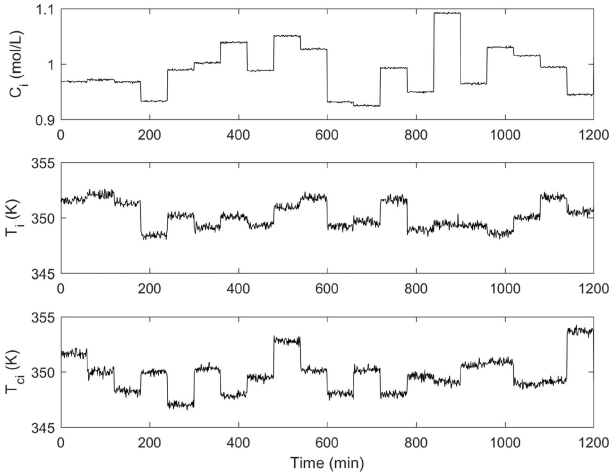


Fig. 10. Example of datasets from CSTR simulation [158].

1) A key challenge in incremental learning for fault diagnosis has been balancing timely FDI with the need to acquire sufficient data for efficient learning. Existing methods typically assume that the system can accumulate a sufficiently rich set of fault samples during operation. However, once a severe fault is detected in practice, the system is usually shut down by operators immediately. In addition, considering that faults often occur internally within the system, it is challenging for experts to provide annotations for fault samples in real time [159]. Most existing methods have overlooked the implementation of the annotation process. Future research could explore intelligent sample selection strategies to prioritize the most informative data for updates, thereby ensuring timely FDI while maintaining learning quality [143], [160].
2) In the existing literature, significant progress has been made in the design of diagnostic model architectures. However, the update strategies of current methods typically do not consider the *temporal relationships* between system measurements, which is a crucial factor in practical applications. In real-life scenarios, industrial processes are continuous in nature. Hence, designing *time-aware* diagnostic model update strategies is of great research significance, especially for incremental-OL-based fault diagnosis methods.
3) As discussed in Section VII, in practical applications, system performance degrades over time. Fault characteristics often vary significantly under different operating conditions, and these conditions can greatly impact system behavior. Existing methods typically assume that the system can remain in a stable process during the online phase. The ILFD diagnostic process is not carefully designed to address performance degradation and operating mode transitions. Accordingly, research on the deployment of active detection and adaptation mechanisms is essential for resolving these challenges and improving system robustness.
4) Although DNN-based ILFD methods have demonstrated excellent performance, their *lack of interpretability* limits the understanding and trust of engineers in the diagnostic results. It is essential to improve the interpretability of diagnostic models, which will greatly facilitate the adoption of these technologies in practical production. In addition, incorporating temporal relationships into the model interpretation process, referred to as dynamic *explainable artificial intelligence* (XAI) techniques, is crucial for understanding state changes in dynamic systems, thereby aiding tasks such as fault localization.
5) In practical industrial environments, terminal computing resources are typically limited, which poses significant challenges for the updating of complex diagnostic models. Existing incremental-offline-learning-based FD approaches usually require substantial computing resources. Under resource-constrained conditions, the effectiveness of model updates may be compromised. By investigating lightweight model architectures and efficient computational methods, it is possible to achieve effective fault diagnosis on terminal devices. Accordingly, the optimization of model performance under resource limitations is essential for ensuring accurate and efficient real-time ILFD.

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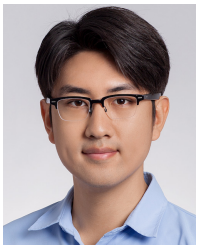
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